# HARSHVARDHAN PATIL

+91 81233 47973 | harshvardhanpatil099@gmail.com | LinkedIn | Portfolio

*Mechanical Design / Aerospace Structures / Systems Engineer /CAD*

## SUMMARY

Aeronautical Engineer with hands-on experience in mechanical design, aerospace structures, CAD assembly development, mechanical packaging, and subsystem integration. Proficient in Catia V5 and SolidWorks, supported by structural validation using FEA. Hands-on experience in structural optimization, carbon-fiber composites, PLM-based configuration control, and strength-to-weight driven design, with exposure to rapid prototyping and design for manufacturing. Academic exposure to propulsion and cryogenic propellant systems complements this experience.

## TECHNICAL SKILLS

**Aerospace CAD Assemblies & Systems Integration:** CATIA V5, Siemens NX, SolidWorks, AutoCAD, Assembly Design, Mechanical Packaging, Interface Definition

**Aerospace Structures & Analysis:** ANSYS, Nastran, HyperMesh, FEA, Structural Validation, Aerospace Structures, Load-Path Definition, Structural Integration, Strength-to-Weight Optimization

**Manufacturing & Engineering Management:** PLM, Configuration Control & Management, DFM, DFAM, FDM Rapid Prototyping, Carbon-Fiber Composites, Engineering Drawings, BOM Preparation, ASME Y14.5 GD&T, ISO 1101, Tolerance Stack-up Analysis

**Programming & Computation:** Python, MATLAB

## PROFESSIONAL EXPERIENCE

**Systems Integration & UAV Design Engineer** **Sept. 2025 – Present**

*Octaknight Labs Pvt. Ltd. · Belagavi*

- Designed aerospace UAV structural components using carbon-fiber composites, applying structural sizing and strength-to-weight optimization under defined load cases.
- Supported UAV airframe development through primary structural architecture, load-path definition, mechanical interfaces, and member sizing for flight, thrust, and landing loads.
- Developed and maintained CAD assemblies while supporting mechanical packaging and subsystem integration within the airframe.
- Performed analytical stress calculations and FEA-based structural validation for structural components under flight and landing loads.
- Supported engineering drawings, BOM preparation, PLM, DFM/DFAM, and FDM rapid prototyping through design and development iterations.

**Mechanical Design Intern** **Oct. 2024 – Nov. 2024**

*Polyflow Precision Pvt. Ltd. · Belagavi*

- Designed machining components considering structural integrity, interfaces, manufacturability, and dimensional control.
- Applied ASME Y14.5 GD&T and tolerance stack-up analysis to improve component fit, interfaces, and manufacturing feasibility.

## PROJECTS

**VTOL Tailsitter Airframe Development**

- Developed the primary structural architecture and CAD assemblies for a VTOL tailsitter airframe, defining load paths and mechanical interfaces for flight and ground load cases.
- Sized structural members and validated performance using analytical calculations and FEA, emphasising strength-to-weight optimisation.
- Evaluated carbon-fiber composite airframe materials and structural failure modes.

**All-Terrain Vehicle — Structural System** / *ASME IAM3D 2026 – Winner*

- Designed an all-terrain vehicle structural system with emphasis on lightweight design and structural integration for multi-terrain operation.
- Developed and optimized CAD assemblies for FDM manufacturing, applying DFM and DFAM principles to component geometry, material usage, and manufacturability.

## ACHIEVEMENTS & LEADERSHIP

- **4× ASME National EFx Winner** — ASME IMECE Brainbolt top-4, IAM3D 2026, IAM3D 2025, SDC 2025, SDC 2026
- **Treasurer** — ASME KLS GIT Student Chapter Feb. 2025 – Jan. 2026

## EDUCATION

**KLS Gogte Institute of Technology** Belagavi, India

B.E. Aeronautical Engineering | CGPA: 7.5 **2022 – 2026**

*Relevant Coursework:* Aircraft Structures, Finite Element Analysis, Fatigue & Fracture Mechanics, Thermodynamics, Fluid Mechanics, Propulsion; *academic exposure:* Cryogenic Propellant Systems (LOX, $LH_2$, $LCH_4$).